Study Questions: Systems (Fiber & Flow)

1. In your own words, describe what makes a skein of yarn a "system" in the Active Inference sense. What are its internal states, external states, and boundary?
2. How does fiber composition (wool vs. cotton vs. acrylic) represent different internal states of a yarn system? Give a specific example of how each fiber type responds differently to the same environmental input.
3. Explain the concept of a Markov blanket using the physical surface of a yarn strand as your example. What passes through this boundary, and what does not?
4. Why do two skeins from different dye lots behave differently even when they share the same label specifications? Connect this to the idea of environmental context shaping a system.
5. How is twist direction (S-twist vs. Z-twist) an internal state variable of the yarn system? What observable behaviors does it generate?
6. A crocheter picks up a skein and squeezes it before buying. What are they doing in Active Inference terms? What are they trying to infer about the system's internal states?
7. How does humidity change the behavior of a wool yarn system? Describe the pathway from environmental state through the boundary to internal state change.
8. Compare a single-ply yarn and a 4-ply yarn as two different systems. How do their internal states (ply structure) lead to different behaviors at the boundary (stitch definition, splitting tendency)?
9. What does the yarn label represent in terms of a generative model? Why might an experienced crocheter trust their hands more than the label?
10. Explain how blocking a finished piece uses environmental manipulation (heat, water, pressure) to change the internal states of a yarn system.

11. How is the concept of "yarn memory" (the tendency of acrylic to return to its original shape after heating) an example of a system maintaining preferred states?
12. Describe the dye lot problem as a systems-level phenomenon. Why can't manufacturers perfectly replicate environmental conditions across batches?
13. How does the concept of fiber "bloom" (the way some yarns puff up after washing) illustrate the relationship between internal states and boundary conditions?
14. A variegated yarn has multiple colors repeating along its length. Is each color section a different system, or is the color pattern part of one system's internal state? Argue your position.
15. How might you think of your yarn stash as a collection of systems? What environmental conditions should you maintain to keep these systems within their viable range?
16. Compare the boundary of a tightly wound cake to the boundary of a loosely wound hank. How does the physical arrangement of yarn change what happens at the system boundary?
17. Why does the same yarn feel different when crocheted at different tensions? Connect this to the idea that the agent (crocheter) changes the conditions at the system boundary.
18. How is superwash treatment on wool an example of modifying a system's boundary to change how it interacts with its environment?
19. Explain why gauge swatching is a form of systems investigation. What are you learning about the yarn system when you swatch?
20. If you were to describe your favorite yarn to someone who has never touched it, what internal states and boundary properties would you communicate? What would be lost without direct sensory contact?